

American Economic History

Week 1 — Introduction

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Today

1. The syllabus: how this class works.
2. Why economic history? Three answers.
3. The most important economic fact ever recorded.
4. Before 1800: the Malthusian world.
5. After 1800: growth accounting, and **A**.
6. What drives **A**? Romer, and a test we can actually run.

The syllabus

How this class works

- **Thematic.** I am not going to teach you the story of American history. We read and argue about papers.
- The real skill: **how to consume cutting-edge empirical research**, and how to make an argument you can sustain with evidence.
- Four things are graded:
 - Oral midterm exam (30%) — Week 7, 19 October
 - Oral final exam (30%) — Monday 14 December, 4:30–7:15pm
 - Research paper (10%) — due 7 December, 11pm
 - Paper presentation (30%) — Weeks 13 and 14

Two things worth saying out loud

- **The oral exams.** Fifteen-minute slots, on Zoom, camera on, laptop not phone — you will use the whiteboard. I tell you two or three topics in advance and I guarantee one of them shows up.
- **AI tools.** Use them. I do. Mason gives you PatriotAI for free. Two conditions:
 - disclose your workflow in the paper, and
 - you sign the result — every citation, every number, every claim.
- **“The AI wrote that part” is not a defense.** The oral exams are where you show, live, that you understand this material.

Why economic history?

Why study economic history? Three answers

1. **To explain the present.** Much of what we see in comparative development today has roots centuries deep.
2. **To inform policy.** History is the only place we can observe how institutions behave over the long run.
3. **To build better theory.** History supplies the variation that lets us discriminate between theories that fit the same modern cross-section.

Nunn (2020), "The historical roots of economic development," *Science*.

1. To explain the present

- Nunn's review is built on a simple observation: **the best predictor of a region's income today is its income a long time ago.**
- The ranking of regional prosperity in 2000 is close to the ranking in 1800. Go back to 1500 and it is still informative.
- **Example.** Dell (2010): the Peruvian *mita*, a forced-labour system that ended in 1812, still shows up in household consumption today. We read this in Week 3.

Today's differences have roots far older than today's policies. That is what makes history load-bearing here.

2. To inform policy

- **Example.** Alsan and Wanamaker (2018) on the Tuskegee syphilis study.
- After disclosure in 1972, older Black men living closer to Tuskegee showed greater medical mistrust, less interaction with physicians, and higher mortality.
- The policy lesson tells you what a public health campaign is up against, and why. We read this in Week 8.

Institutions leave residues that outlive the institution. Policy that ignores the residue misfires.

3. To build better theory

- Theories of urban location come in two flavours: cities sit where they sit because of **locational fundamentals** (rivers, harbours, coal), or because of **increasing returns and path dependence**.
- Both fit the modern cross-section. They differ in what happens after a **large temporary shock**.
- **Example.** Davis and Weinstein (2002) use the Allied bombing of Japanese cities. City sizes return to trend within about twenty years.
- That is a fact about theory, and only history could produce it. Week 6.

The great divergence

The most important economic fact ever recorded

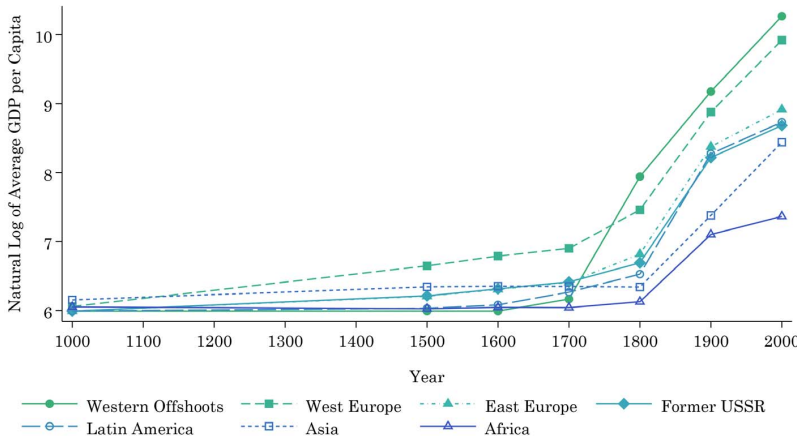


Fig. 1. Relative economic prosperity of different regions of the world from 1000 to 2000 CE.

A divergence in two senses

1. **Over time.** For almost all of recorded history, income per person went essentially nowhere. Then, around 1800, it went up — and kept going.
2. **Across regions.** The take-off did not happen everywhere at once. Western Europe and the Western Offshoots left first; others left later, or barely.

Today the richest *countries* are roughly **forty times** as rich as the poorest. In 1500 the richest *region* was perhaps **twice** the poorest. (The figure plots regional averages, where today's gap is nearer twenty-fold.)

Why?

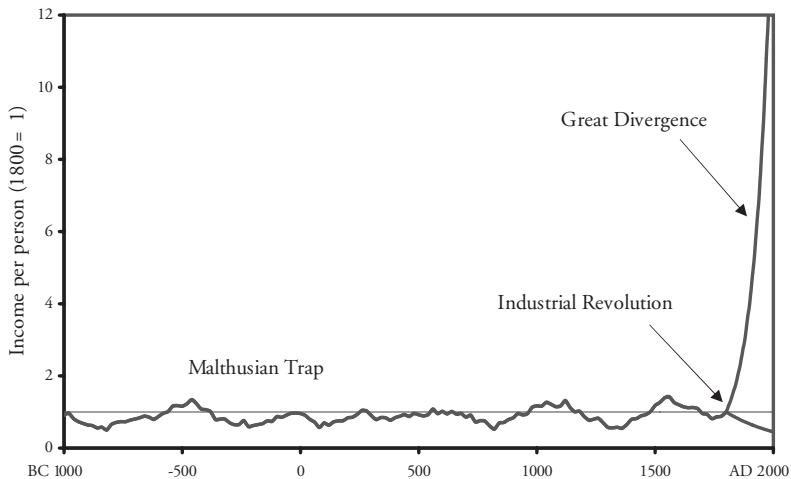


Figure 1.1 World economic history in one picture. Incomes rose sharply in many countries after 1800 but declined in others.

Clark, *A Farewell to Alms* (2007), Figure 1.1. The same fact, drawn as one picture.

Before 1800: the Malthusian world

Two worlds, one hinge

- Before 1800, essentially everywhere, the world looks **Malthusian**: technology improves, population grows, income per person does not.
- After 1800, with staggered timing, regions start to look **Solovian**: sustained growth in income per person.
- So we need two models, and an account of the hinge between them.

The version of Malthus I will draw is Greg Clark's, from *A Farewell to Alms*. Three curves: a technology schedule, a fertility schedule, a mortality schedule.

Two assumptions

Assumption 1: a fixed factor

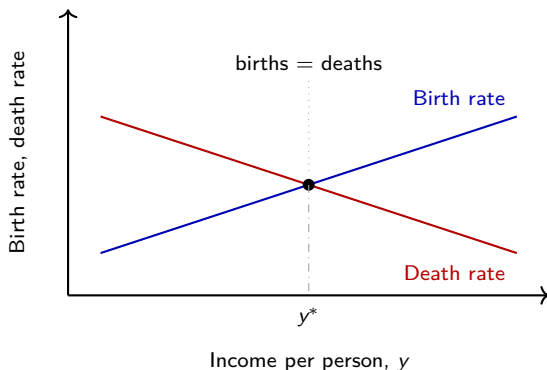
Production uses a factor that cannot be expanded — conventionally **land**. More honestly: **how widely you can trade for food and fuel**. Fixed in the short run, so output per person falls as population rises.

Assumption 2: fertility rises with income

When material living standards rise, people marry earlier and have more surviving children. Birth rates go **up** with income; death rates go **down**.

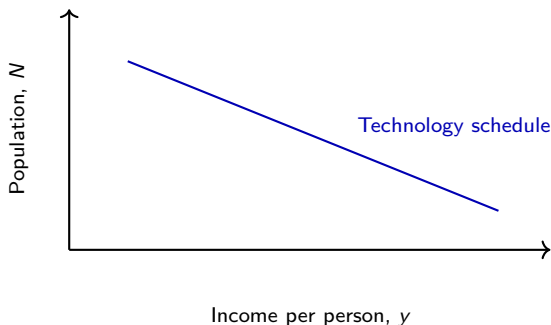
Everything that follows is a consequence of these two assumptions. When we get to 1800, ask which one broke.

Birth and death rates pin down subsistence income



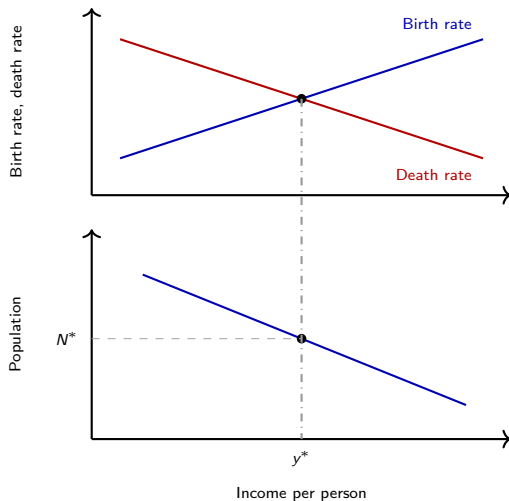
- Right of y^* : births exceed deaths, so **population grows**. Left of it, population shrinks.
- **Subsistence income y^* is a property of these two schedules** — not of biology, and **not of technology**.

The technology schedule — drawn against income



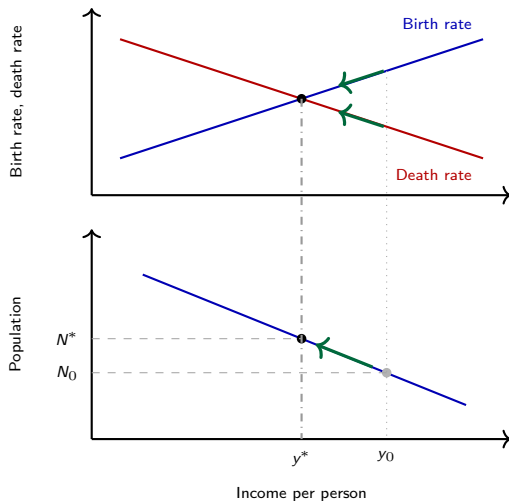
- Same fact as before, drawn the other way round. Each point pairs a **population** with the **income** that population can achieve, given the technology and the fixed factor.
- More people, less income each. So it slopes **down**.
- Its **position** is what technology is — a better plough, a better rotation, a wider trading network all shift it **out**.

Long-run equilibrium



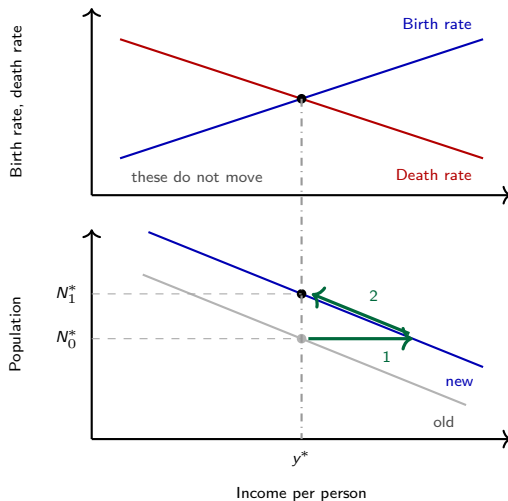
The top panel fixes **income** at y^* . Drop that same y^* into the bottom panel and it fixes **population** at N^* .

Out of equilibrium



Start at y_0 , richer than subsistence. Births beat deaths, population grows, and the economy **slides back up the schedule to y^*** — ending with **more people, and income back down at subsistence**.

An isolated technological advance



1. At the old population, better technology means more income.
2. Births beat deaths, so population grows until income is back at y^* . The vertical line **never moved**.

The Malthusian trap

The trap. A technological improvement raises income *temporarily*. In the long run income returns to y^* and **only population density is permanently higher**.

This is a useful result, in two directions.

- 1. It explains a puzzle.** Mokyr's *The Lever of Riches* documents millennia of genuine advance in **technique**. The Malthusian model explains why none of it made anyone richer for long: it raised A , and the fertility schedule converted the gain into people.
- 2. It tells us how to measure.** In a Malthusian world, better technology and better institutions show up as **higher population density**, while income per person stays pinned at y^* .

Why point (2) matters for this course

- If you are looking for evidence that some pre-1800 society was doing something right, **do not look at GDP per capita**. It will be approximately y^* everywhere.
- Look at **how many people the place supported**, and at real wages relative to their own history.
- This is one reason historical demography is not a niche interest in this field. Population is the outcome variable that moves.

A flat income series before 1800 is what the model predicts when a great deal happened.

After 1800: where does growth come from?

We need a different model

- After 1800 income per person grows, and keeps growing. Something in the Malthusian setup has broken.
- I am not going to derive Solow. Instead, take the production function and ask a narrower question: **where did the growth come from?**

$$Y = A K^{\alpha} N^{\beta} T^{\gamma}$$

Y output, K capital, N labour, T land, A total factor productivity.

We can **measure** K , N , and T . We cannot measure A .

Labour is the same N we used for population — in a Malthusian world they are the same thing, and that will matter again before the end of today.

What is A ?

$$A = \frac{Y}{K^\alpha N^\beta T^\gamma}$$

- A is **what is left over** once we have measured everything we know how to measure.
- It is a residual.

Moses Abramovitz called it **“the measure of our ignorance.”**

Which would be a footnote, except that it turns out to be most of the answer.

Growth accounting

Take logs:

$$\ln Y = \ln A + \alpha \ln K + \beta \ln N + \gamma \ln T$$

Differentiate with respect to time:

$$\underbrace{\frac{\dot{Y}}{Y}}_{\text{output growth}} = \underbrace{\frac{\dot{A}}{A}}_{\text{TFP growth}} + \alpha \underbrace{\frac{\dot{K}}{K}}_{\text{capital}} + \beta \underbrace{\frac{\dot{N}}{N}}_{\text{labour}} + \gamma \underbrace{\frac{\dot{T}}{T}}_{\text{land}}$$

- Everything is now in **percentage changes**, and the pieces add up.
- Rearranged, $\dot{A}/A$ is whatever output growth is not explained by measured input growth.

Extensive versus intensive growth

Extensive growth

Output rises because **inputs** rise. More workers, more land under plough, more capital. Output per person need not change at all.

Intensive growth

Output rises **per unit of input**. This is $\dot{A}/A$. It is the kind that *can* make people richer — though in a Malthusian world it does not, because population absorbs it.

The Malthusian world can deliver plenty of extensive growth. It cannot deliver sustained intensive growth — and if it did, the fertility schedule would eat it.

Where does the United States sit?

U.S. in	GNP per capita (1989 \$)	Which puts it, in 1989, next to
1710	550	above Bangladesh (180), below Angola (610)
1775	750	Philippines (710), Dominican Republic (790)
1840	1,350	Thailand (1,220), Turkey (1,370)
1880	2,650	Uruguay (2,620), Yugoslavia (2,920)
1929	6,150	Saudi Arabia (6,020)
1945	10,450	Singapore (10,450)
1960	11,100	New Zealand (12,070)
1989	20,910	sixth in the world

- **Not** the poorest. At \$550 the colonies sat around **83rd of 124** — three times Bangladesh, seven times Mozambique. Atack and Passell are explicit: the colonists “were not desperately poor in the way that the citizens of Ethiopia, Bangladesh, or Haiti are today.”
- What the table shows is the **distance travelled**: from the range of today's poorer countries to **sixth in the world**, in under three centuries.

Selected rows from Atack and Passell, *A New Economic View of American History*, Table 1.2. Their source: World Bank, *World Development Report 1991*.

Accounting for US growth, 1840–1990

TABLE 1.5

Accounting for Growth, 1840–1990

PANEL A

Annual Rate of Growth of:

Period	Labor	Capital	Land	Net National Product
1840–1860	3.42%	6.57%	3.73%	4.75%
1870–1930	2.24	4.35	2.55	3.75
1940–1990	1.59	3.14	0.34	3.22

PANEL B

Percentage of the Growth in Net National Product Attributable to:

Period	Labor	Capital	Land	Productivity (the residual)
1840–1860	49%	26%	10%	15%
1870–1930	43	27	4	27
1940–1990	41	14	0	45

Source: After Edward F. Denison, *Sources of Economic Growth in the United States* (Washington D.C.: Brookings Institution, 1962) and Lance Davis et al., *American Economic Growth* (New York: Harper and Row, 1972): 34–39. Copyright © 1972 by Harper & Row Publishers, Inc. Reprinted by permission of HarperCollins Publishers, Inc.

Read Panel B again

- 1840–1860: productivity is **15%** of output growth. Labour is 49%.
- 1870–1930: productivity is **27%**.
- 1940–1990: productivity is **45%** — and land contributes **zero**.

Over time the share of American growth coming from **the residual** rises until it is the largest single component. **Most of modern growth is *A*** — the one term we cannot measure directly.

And we said *A* is the measure of our ignorance.

Or is it?

What drives A ? Romer, and a test

Ideas are not like other inputs

- Romer's starting point: an idea is **non-rival in consumption**.
- If I use a hammer, you cannot use that hammer. If I use the Pythagorean theorem, you can use it at the same moment, at no cost to me.
- Ideas are also **partially non-excludable**: hard to stop others using them once they exist.

Non-rivalry

One person's use of an idea does not diminish anyone else's ability to use it. Capital and land are rival. Ideas are not.

Why non-rivalry makes population special

- Suppose each person has some small probability of having a good idea in their lifetime.
- Because the idea is non-rival, **everyone** can use it once it exists — not just the person who had it.
- So more people means more ideas, **and** the ideas are shared across the whole population.

If Romer is right, the level of **population** should drive the growth rate of **technology**.

Now put Romer inside the Malthusian model

Two statements, each from one of our two models:

From Romer. Each person has some small chance p of an idea, and researchers stand on the shoulders of the existing stock, so $\dot{A} = p A N$ and therefore

$$\frac{\dot{A}}{A} = p N \propto N$$

From Malthus. Income sits at y^* , so any gain in A is converted into people. If population adjusts fast relative to a millennium, the rate of population growth tracks the rate of technological change:

$$\frac{\dot{N}}{N} \propto \frac{\dot{A}}{A}$$

Combine them:

$$\boxed{\frac{\dot{N}}{N} \propto N}$$

Why that equation is worth something

- $\dot{N}/N \propto N$ is true only if **both** things hold: endogenous growth *and* a Malthusian world.
- It is a statement about **population** alone — and population is the one series we have for the very long run.
- So it is testable, over a period no other economic hypothesis can reach.
- One assumption is doing real work: that population adjusts *quickly* to income. Over centuries that is defensible — as our out-of-equilibrium frame showed, adjustment is not instant. Kremer says so openly.

Kremer (1993) ran the test: world population, one million BC to 1990. Plot the growth rate against the level.

Kremer's test

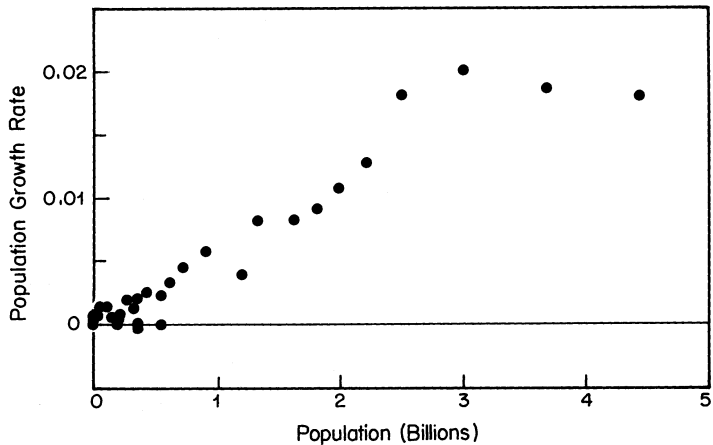


FIGURE I
Population Growth Versus Population

What the picture shows

- For almost the entire record the relationship is a **straight line through the origin** — exactly $\dot{N}/N \propto N$, across a million years.
- Then it **goes flat**. But read the axis: the break comes at roughly **three billion people** — the world of the **1960s**, a century and a half after the Industrial Revolution began.
- That is **assumption 2 failing**. Income stops converting into births. The **demographic transition**, arriving at world scale.

This is the strongest evidence we have that ideas are non-rival, and that endogenous growth explains A — the thing that drives modern growth.

Two dates, one exit

- **Around 1800:** income per person takes off, in Britain and the Western Offshoots first. That is the date in Clark's picture and Nunn's.
- **Around 1960:** the fertility link finally breaks in Kremer's picture.
- The century and a half between them is the **staggered exit**. The leaders left the Malthusian world early; the world as a whole stayed close to Malthusian until most of humanity had followed.

Kremer's plot is a world aggregate, so it is dominated by wherever most people live. It breaks late for the same reason the divergence has a second sense: the exit did not happen everywhere at once.

Where this leaves us

What we just did

- **Theory** gave us two models and a combined prediction.
- **History** gave us a period long enough for the prediction to have something to say.
- **Data** — a population series running back a million years — let us actually look.

That combination is what this field is. **Not one of the three on its own.**

And notice what Kremer actually did: two off-the-shelf models, one substitution, and a claim you can test against a million years of data. **It is one of the most impressive pieces of economics I know.**

What the picture does not tell us

- The pictures **date** the exit: it begins around 1800 among the leaders (Clark, Nunn) and completes at world scale in the twentieth century (Kremer).
- They do not say **why it began then** — and not in 1500, or 2100.
- And they do not say **why it began there** — northwest Europe and its offshoots first, everyone else late or, in income terms, not yet. Remember the second sense of the divergence.

Non-rival ideas explain why sustained growth is **possible**. They do not explain its **timing** or its **geography**. That is the open question this course is about.

Why then, and why there? (Weeks 3–5)

- **Week 3, Colonialism.** Institutions imposed centuries before 1800 sorted which places could *use* the new technology when it arrived (Acemoglu et al.), and extraction leaves scars measurable today (Dell). Conley and Kelly: and how sure can we be of claims shaped like that?
- **Week 4, Institutions and culture.** What makes a state strong enough to support markets and constrained enough not to eat them — and where the preferences themselves come from (Bazzi et al.).
- **Week 5, Industrialization.** The direct attack. Allen: invention responded to *prices* — high wages and cheap coal made the machines pay in Britain and nowhere else. Hanlon: causal proof that invention responds to input supplies. Clark: the machines travelled; **the productivity did not**. Whatever *A* is, it is not just blueprints.

What keeps the engine running — and who feeds it? (Weeks 6–11)

- **Cities (6).** Romer's mechanism has an address: density is where ideas get exchanged.
- **Health (8).** Minds are the input. The disease environment — and trust in medicine — decide who participates.
- **Infrastructure (9).** “Land” was never acreage; it was how widely you can trade. Railroads are assumption 1 dissolving in real time.
- **Slavery (10).** The counterexample inside the success story: institutions built to *exclude*. When ideas are non-rival, locking people out of schooling and markets shrinks the idea pool for everyone.
- **The Depression (11).** Modern growth is not self-running. The machinery seized once, within living memory of your grandparents.

Every week is a piece of the same question: **what makes A grow, and who gets to use it.**

One more thing

- Kremer's engine runs on **minds**: $\dot{A}/A \propto N$.
- **Week 12.** Bloom, Jones, Van Reenen, and Webb: we now put more than twenty times the researchers on the problem that the 1930s did, for the same growth rate. **Ideas may be getting harder to find.**
- And Jones (2026), published this summer: what happens if minds stop being the binding input — if machines can have ideas too?

Around 1800 the idea production function was rewritten once.

You are alive for the second rewrite.

Weeks 2 through 11 are about the first one. Week 12 is about yours.

Next week

- **Week 2 (31 August): Measurement, Causality & New Data.**
- Read Freedman first — “shoe leather.” Then the two Bellemare primers. Abramitzky et al. shows where the field's data now come from.

How do we know any of what I claimed today? That is next week's question.